ECON484 Machine Learning
Midterm Exam
04.05.2026

Student Number	
Name & Surname	

Instructions:

- This exam is in **two parts**. The first part of the exam is **closed resource**. The questions in this part are really simple and you should not need any assistance in that part. The second part of the exam is **open resource.** A resource could be anything from your lecture notes to Artificial Intelligence tools. However, your classmates and other persons are not resources.
- In the first part you will have some simple calculated results by hand, and in the second part you will have to reflect on those results. You may be making some mistakes in the first part. You may even be expected to have mistakes. This is by design, so that you can reflect on those mistakes in the second part.
- **Do not spend excessive time to re-check Part 1.**

Total exam duration is **75** minutes. You are not allowed to use your resources in the first **30** minutes to cover Part 1.

Question	**Points**
1	____ / 5
2	____ / 20
3	____ / 25
4	____ / 15
5	____ / 10
6	____ / 15
7	____ / 10
Total	____ / 100

PART 1 – CLOSED RESOURCE (50 Points)

QUESTION 1 (Seeds, Random Numbers and Shuffling; 5 points)

In this question you will create your unique random number seed that you will use in this exam. **Based on the seed, you will have different computational results than your classmates.**

Follow this procedure:

- First add the digits on you student number (ie. 1234567 means 1+2+3+4+5+6+7=28)
- Then find the remainder of this number when divided by 5 (ie. 28 divided by 5 is 5 with remainder 3)
- The seed will be this remainder. (ie. seed is 3)

Calculations: **Seed:**

Now you can read the case study.

CASE STUDY

You were in your favorite coffee shop with your classmate from ECON484 and discussing experiment ledgers and how they could be used when trying to predict popular indexes such as stock market or USD. Then someone interrupts you: An older man with a grey goatee, dark circles around his eyes an wide smile.

He introduces himself. Apparently he was a big time stock broker back in the 80s and 90s when the markets were roaring. He says their generation never applied complex algorithms, or methodical work like they do now. Nevertheless their generation made tenfold more money. His view is that the more complex things get, the more mistakes are made.

You like his approach and ask him, how he would track USD back in the day. He says he had a simple average-based estimation algorithm, and based on the algorithm he would not decide on the exact amount, but the up or down behavior of the value. This makes some sense into you. Most modern estimation algorithms are modified versions of a moving average anyway. The idea of up versus down also makes some sense. Most investors would not bother themselves with how much as long as they are in the winning side. So you ask more about this algorithm.

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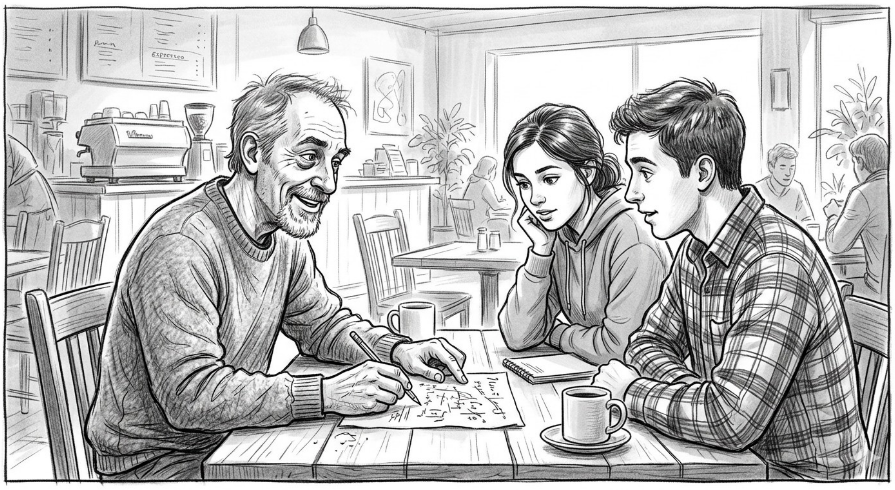

Figure 1: You listening to a winning strategy from the 90s.

His algorithm is as follows:

1. Keep the past 25 trading days data at hand. I would have the actual value, and if it was “up” or “down”
2. Pick 5 random days from this data. I used to shuffle the data set, and pick the last 5 days in the shuffled deck, but you could pick a different approach.
3. Those 5 random days have a vote (ie. 3 ups and 2 downs means an “up”)
4. You select the result of the vote as your position.

Because you are a Gen-Z you do not take his word for granted and like to take this algorithm for a spin. You decide to:

1. Collect USD Forex data for a period of 25 business days and construct a table
2. Use your own seed to shuffle the data set as instructed and pick the five days
3. Estimate whether it is up or down in the next business day.

Day	March 2	March 3	March 4	March 5	March 6
USD	43,8850	43,8913	43,8932	43,9135	43,9055
Vote	UP	UP	UP	UP	DOWN
Day	March 9	March 10	March 11	March 12	March 13
USD	44,0007	43,9826	43,9964	44,0310	44,0210
Day	UP	DOWN	DOWN	UP	DOWN
Day	March 16	March 17	March 18	March 24	March 25
USD	44,1060	44,1207	44,1325	44,2636	44,2737
Vote	UP	UP	UP	UP	UP
Day	March 26	March 27	March 30	March 31	April 1
USD	44,2828	44,2887	44,3841	44,3961	44,3938
Vote	UP	UP	UP	UP	DOWN
Day	April 2	April 3	April 6	April 7	April 8
USD	44,4155	44,4306	44,5085	44,5255	44,4527
Vote	UP	DOWN	UP	UP	DOWN

Table 1. USD Forex Values and Their “Votes”

You also note that **on April 9th, the USD Forex Buying value was 44,5102 which means an "up" ticker**.

Then you ask your favorite AI agent to create a simple Python script to shuffle the dates as instructed and the fun begins.

QUESTION 2 (Using Shuffled Results; 20 points)

When you use your AI assistant to create some code, you also **execute the code** and get results (you can check the appendix if you'd like, but no need for this question). Here are the results, in a nice tabular form for you.

Seed	Shuffled List of Days
0	['March 25', 'March 2', 'April 7', 'March 31', 'March 10', 'March 9', 'March 11', 'March 16', 'April 6', 'April 2', 'March 6', 'March 4', 'April 3', 'March 5', 'March 30', 'March 17', 'April 1', 'March 13', 'April 8', 'March 26', 'March 27', 'March 12', 'March 3', 'March 24', 'March 18']
1	['March 16', 'April 2', 'April 8', 'April 3', 'April 7', 'March 9', 'April 6', 'March 24', 'March 27', 'March 13', 'March 17', 'March 30', 'March 2', 'March 11', 'March 3', 'March 10', 'March 18', 'April 1', 'March 25', 'March 26', 'March 5', 'March 12', 'March 4', 'March 31', 'March 6']
2	['March 5', 'April 1', 'March 27', 'March 18', 'March 2', 'April 6', 'March 6', 'March 11', 'March 24', 'March 26', 'March 31', 'April 3', 'March 25', 'March 16', 'March 30', 'April 2', 'April 8', 'March 10', 'March 12', 'March 13', 'March 9', 'March 17', 'April 7', 'March 4', 'March 3']
3	['March 10', 'March 27', 'April 6', 'March 25', 'March 3', 'March 9', 'April 2', 'March 16', 'March 13', 'March 24', 'April 8', 'March 18', 'March 5', 'March 12', 'April 3', 'April 7', 'March 2', 'March 4', 'March 26', 'April 1', 'March 17', 'March 6', 'March 30', 'March 31', 'March 11']
4	['April 8', 'March 16', 'March 24', 'March 10', 'March 17', 'March 3', 'March 30', 'March 9', 'April 7', 'March 25', 'April 6', 'March 27', 'April 2', 'April 1', 'March 12', 'April 3', 'March 2', 'March 31', 'March 4', 'March 6', 'March 26', 'March 18', 'March 5', 'March 13', 'March 11']

Table 2. Seed-Based Shuffle Results

a) (10 points) Based on you **personal unique seed** in Question 1, **select your shuffle results (ie. circle a seed in Table 2) and then pick the last 5 days from Table 2**. Then based on your selected five days from Table 2, **fill out the following table using data from Table 1**.

	Day 1	Day 2	Day 3	Day 4	Day 5
Actual Day					
USD Value					
Vote					

Table 3. Voting for April 9th

b) (5 points) Now hold the vote based on Table 3 and predict if it is an "up" or "down" for April 9th.

c) (5 points) Does you prediction hold?

QUESTION 3 (Ledgers, Understanding ML Process, Reproducibility, Metrics; 25 points)
In this question you are given very simple and straightforward questions that ask you to translate this case into machine learning terminology (ie. Talk the talk).

a) (5 points) You ask yourself and your classmate about use of ledgers for this algorithm. What type of data would you need to store in the ledger for this type of up/down estimation?

b) (5 points) If this is an ML based algorithm, what would be the “features” in this algorithm?

c) (5 points) If this is an ML based algorithm, what would be the “output” in this algorithm?

d) (5 points) If your classmate uses the same student number and the same Table 2 as you, will they get a different result? Why or why not?

e) (5 points) Looking at Table 1 what is the frequency of “up” and “down” votes in the 25 day sample? Then calculate the frequency of “up” and “down” votes, but only for the days where the USD value was above 44,10000. Please show your work in detail (ie. which days are selected). How does this selective frequency compare to the overall sample frequency? What does this tell you about the model's potential bias?

PART 2 – OPEN RESOURCE (50 Points)

In Part 2, you may use external resources, including AI tools. However:

- The questions are intentionally structured so that generic prompts (e.g., “give me the correct answer”) will not be sufficient.
- You are responsible for verifying and understanding any output you use.
- Answers will be graded primarily on reasoning and clarity, not just the final result.
- If you use AI, briefly indicate how it assisted you (e.g., idea generation, structuring, verification).

Submissions that demonstrate clear understanding will receive full credit, even if intermediate results from Part 1 contain errors.

QUESTION 4 (Assessment of Model Fragility; 15 points)
a) (10 points) This model predicts a single day, however you could easily back-test for a longer period. Describe the procedure for back-testing for the whole month of April. You should still stick to the random shuffling concept in you procedure.

b) (5 points) Eventually the algorithm would fail on some days. For those days, does the algorithm fail because of the logic, or because the sample size (n=5) is too small to capture market volatility?

QUESTION 5 (Drift; 10 points)
a) (5 points) Calculate the average USD value for the first 5 days and the last 5 days of Table 1. Use these two specific averages to justify whether the Broker’s 90s algorithm is experiencing Data Drift in 2026. A generic answer without these two calculated values will receive zero credit.

b) (5 points) The Broker claims this worked in the 90s. Based on the 2026 data in Table 1, identify signs of Concept Drift that make this “simple” strategy dangerous today.

QUESTION 6 (Data Leakage and Validation; 15 points)

a) (5 points) Explain if and why the Broker's "random shuffling" method might lead to Data Leakage.

b) (10 points) Propose a walk-forward validation structure using the dates in Table 1 that would be more scientifically sound. If needed you can also draw assistive diagrams to describe how you handle the data set.

QUESTION 7 (Short Reflection Questions; 10 points)

a) (5 points) The "Up/Down" vote is a primitive form of data encoding. Turning continuous USD values into binary votes loses information. **What specific information are we losing** by encoding the data this way, and **how would a weighting feature change the model's sensitivity?**

b) (5 points) If you developed Python code to execute this algorithm (with or without AI assistance), and you wanted an automated process with the program automatically writing into the ledger, you'd have to take into account ledger specific issues as well. AI assistance in this regard gives a standard list: seeds, versioning, timestamps. Let's **introduce a constraint.** Assume your ledger has **a storage limit of only 3 fields per entry**, and you are designing the ledger part of shuffling (ie. Appendix 1). Which 3 specific pieces of metadata are mechanically mandatory to reproduce the shuffle results in Table 2? Justify your decision.

Appendix 1. Sample Python code to shuffle the dates.

```
import random

# The list of dates provided
dates = [
    "March 2", "March 3", "March 4", "March 5", "March 6",
    "March 9", "March 10", "March 11", "March 12", "March 13",
    "March 16", "March 17", "March 18", "March 24", "March 25",
    "March 26", "March 27", "March 30", "March 31", "April 1",
    "April 2", "April 3", "April 6", "April 7", "April 8"
]

seeds = [0, 1, 2, 3, 4]

for seed in seeds:
    # Work on a copy to keep the original list intact for each iteration
    shuffled_dates = dates.copy()

    # Initialize the random number generator with the specific seed
    random.seed(seed)

    # Perform the in-place shuffle
    random.shuffle(shuffled_dates)

    print(f"--- Seed {seed} ---")
    print(shuffled_dates)
    print("\n")
```